

Algebra 1

Unit 13

Lesson H2 Homework

Name _____

Date _____

Period _____

Use the functions $k(x) = 3x^2 + 4x - 5$ and $p(x) = 6 - 2x$ to determine each new function. Write your answer in standard form.

1. $(k + p)(x)$

2. $(k - p)(x)$

3. $(p - k)(x)$

4. $(k \cdot p)(x)$

5. $\left(\frac{p}{p}\right)x$

6. Write the restriction on the domain for $\left(\frac{p}{p}\right)x$.

Juan and Marina are renting a car for a cross country road trip. The car costs \$155 per day. The rental includes 100 miles for free and then charges \$0.25 per mile. The function $C(m) = 0.25(m - 100) + 155$ models the cost of the car, C , in terms of miles traveled, m . They determine gas will cost \$1.15 for each mile they travel. The company charges a \$40 fee for the first tank of gas. The function $G(m) = 1.15m + 40$ models the cost of gasoline for their vacation.

7. What is a reasonable domain for $C(m)$?

8. What is a reasonable domain for $G(m)$?

9. Determine the function that gives the total cost, $T(m)$, in dollars, based on the miles traveled.

10. What is a reasonable domain for $T(m)$?